05 - Text + Image Fusion

Goal: combine the text and image models and check if this beats text alone.

1. Turn the text SVM scores into probabilities.
2. Average text and image probabilities: `w x text + (1 − w) x image`.
3. Compare text alone, image alone and fusion on the same 16,984 validation products.

1. Text probabilities

We use the optimised text model from notebook 04 (all features, binary TF-IDF, LinearSVC with C=1). A LinearSVC gives scores, not probabilities, so we wrap it in `CalibratedClassifierCV`. It learns on train (3-fold CV) how to turn the scores into probabilities.

```
Text probabilities saved: (16984, 28)
Each row sums to 1: True
```

2. Match text and image on the same products

```
Products scored by both: 16,984
```

3. Choose the text weight `w`

We try `w` from 0 (image only) to 1 (text only).

Limitation: image probabilities exist only for validation, so `w` is chosen on validation. The fusion score is therefore slightly optimistic.

```
w_text  weighted_f1
   0.0       0.4817
   0.1       0.5446
   0.2       0.6434
   0.3       0.7453
   0.4       0.8095
   0.5       0.8427
   0.6       0.8492
   0.7       0.8501
   0.8       0.8476
   0.9       0.8453
   1.0       0.8430

Best text weight: 0.7
```

4. Text vs image vs fusion

	model	weighted_f1	macro_f1
0	Text only	0.8430	0.8326
1	Image only	0.4817	0.4230
2	Fusion (w=0.7)	0.8501	0.8401

	code	category	text_f1	fusion_f1	gain
0	2705	New Books	0.711	0.773	0.062
1	10	Used Books	0.566	0.610	0.044
2	40	Games	0.753	0.781	0.028
3	2403	Book Lots	0.768	0.791	0.023
4	2280	Magazines	0.810	0.826	0.016
5	2462	Used Games	0.825	0.838	0.013
6	1160	Cards	0.954	0.965	0.011
7	1140	Figurines	0.830	0.839	0.009
8	1940	Food	0.927	0.936	0.009
9	1281	Board Games	0.616	0.625	0.009
10	2582	Garden Furniture	0.768	0.775	0.006
11	50	Game Accessories	0.836	0.841	0.005
12	2585	Garden Tools	0.833	0.837	0.004
13	2522	Stationery	0.944	0.946	0.002
14	1560	Furniture	0.852	0.852	0.000
15	1301	Baby Socks	0.972	0.972	0.000
16	2905	PC Games	0.988	0.988	0.000
17	2220	Pets	0.883	0.883	-0.001
18	1302	Outdoor Toys	0.837	0.836	-0.001
19	2060	Decor	0.828	0.827	-0.001
20	1320	Baby	0.862	0.860	-0.002
21	2583	Pool	0.981	0.978	-0.003
22	60	Consoles	0.879	0.876	-0.003
23	1920	Textiles	0.920	0.917	-0.003
24	1280	Toys	0.712	0.707	-0.005
25	1300	RC Models	0.926	0.919	-0.007
26	1180	Wargames	0.697	0.685	-0.012

Analysis

- Text alone (calibrated): weighted F1 0.8430. Image alone: 0.4817.
- Fusion with 70% text and 30% image: weighted F1 0.8501 (+0.7 points) and macro F1 0.8401 (+0.8 points).
- The weight matters little between 0.6 and 0.8 (0.8476 to 0.8501).
- Book categories gain the most: New Books +6.2 points, Used Books +4.4, Book Lots +2.3, Magazines +1.6. These are the categories where text is weakest.
- No category loses much. The largest loss is Wargames (-1.2 points).
- Limitation: `w` was chosen on validation, so the gain is slightly optimistic. A clean version needs image predictions on train to choose `w` with cross-validation.

Conclusion

Combining text and image gives the best model so far: 0.8501 weighted F1 on validation, against 0.8430 for text alone. Images help mainly on book categories.